# Intent Classification in Banking Customer Chatbot Interactions: A Comparative Study of Traditional and Deep Learning Approaches

**Universidad Panamericana**
NLP – Text Classification

Ethan Leonel Garcia Melena

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**Abstract**

This paper presents a comparative study of five machine learning models for intent classification on the BANKING77 dataset, a benchmark corpus of 13,083 customer banking queries annotated with 77 distinct intent categories. We evaluate three traditional machine learning approaches using TF-IDF features — Logistic Regression, Support Vector Machine (LinearSVC), and Random Forest — alongside two deep learning models: a Bidirectional LSTM with a GloVe-initialized embedding layer and an additive attention mechanism, and a fine-tuned DistilBERT transformer. All models are evaluated on accuracy, precision, recall, macro F1, and weighted F1. Results show that DistilBERT achieves the best performance across all metrics (Accuracy: 0.932, F1 Macro: 0.932), followed by the Bidirectional LSTM with GloVe embeddings (F1 Macro: 0.899) and SVM (F1 Macro: 0.894). These findings highlight both the superiority of pre-trained transformer models on fine-grained intent classification and the measurable benefit of pre-trained static word vectors over random initialization in recurrent architectures.

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# 1 Introduction

The automation of customer service has become a strategic priority for financial institutions, driven by the need to reduce operational costs and improve response times. Natural Language Processing (NLP) plays a central role in this transformation through the task of *intent classification*: given a user utterance, determine what the user wants to accomplish (Liu & Lane, 2016). In the banking domain, this capability is essential for routing queries to the appropriate automated response or human agent, enabling conversational AI systems to handle requests ranging from card activation to fraud reporting.

Intent classification in banking is particularly challenging due to the high semantic similarity between many intent categories. For example, distinguishing between *card_not_working* and *card_payment_wrong_exchange_rate* requires understanding fine-grained contextual differences that simple keyword matching cannot capture. This motivates the use of both traditional and deep learning approaches, where the latter can leverage contextual embeddings to better model these distinctions.

This project evaluates five models on the BANKING77 dataset (Casanueva et al., 2020), a standard benchmark for intent detection in the banking domain. The models span a spectrum from classical statistical methods — Logistic Regression, SVM, and Random Forest with TF-IDF features — to neural architectures including a Bidirectional LSTM with GloVe pre-trained embeddings (Pennington et al., 2014) and a fine-tuned DistilBERT (Sanh et al., 2019) transformer. The primary objectives of this study are:

- To benchmark classical and neural models on a fine-grained, 77-class intent classification task.
- To quantify the performance gap between traditional feature engineering (TF-IDF) and pre-trained representations (GloVe, DistilBERT).
- To assess the impact of replacing random embedding initialization with GloVe pre-trained vectors in a BiLSTM architecture.
- To analyze per-class performance to identify which intent categories are systematically harder to classify.

The remainder of this paper is organized as follows. Section 2 reviews related work on intent detection. Section 3 describes the dataset, preprocessing pipeline, and model architectures. Section 4 presents experimental results and comparisons. Section 5 analyzes findings and discusses limitations. Section 6 concludes the paper.

# 2 Related Work

Intent classification is a well-studied problem in task-oriented dialogue systems. Early approaches relied on bag-of-words representations combined with linear classifiers such as Support Vector Machines (Joachims, 1998), which proved effective for text categorization tasks with clear lexical signals. These methods remain competitive baselines in many NLP benchmarks due to their computational efficiency and interpretability.

With the rise of deep learning, recurrent neural networks — particularly LSTMs (Hochreiter & Schmidhuber, 1997) — became dominant for sequence classification tasks. Their ability to model temporal dependencies in text made them particularly suited for understanding the structure of natural language utterances. Liu & Lane (2016) demonstrated that incorporating attention mechanisms into BiLSTM architectures leads to significant improvements in intent detection by allowing the model to focus on the most discriminative tokens in a query.

The development of pre-trained static word embeddings represented an early but significant transfer learning approach in NLP. Pennington et al. (2014) proposed GloVe (Global Vectors for Word Representation), which encodes global co-occurrence statistics from large corpora into dense vector representations. When used to initialize recurrent model embedding layers, GloVe vectors provide a meaningful starting point that reduces the burden of learning word semantics from scratch, particularly in low-to-medium data regimes.

The introduction of transformer-based pre-trained language models marked a paradigm shift in NLP. Devlin et al. (2019) showed that BERT, pre-trained on large corpora and fine-tuned on downstream tasks, achieves state-of-the-art performance across numerous benchmarks. Sanh et al. (2019) proposed DistilBERT, a distilled version of BERT that retains 97% of BERT's performance while being 40% smaller and 60% faster, making it more practical for deployment.

For the specific task of banking intent classification, Casanueva et al. (2020) introduced the BANKING77 dataset and established baseline results using several embedding-based approaches. They found that USE (Universal Sentence Encoder) and ConveRT achieved strong performance, with BERT-based models outperforming shallower methods. Our work extends this line of inquiry by directly comparing TF-IDF-based classical models against deep learning architectures under identical experimental conditions, and by quantifying the specific benefit of GloVe initialization in the BiLSTM setting.

# 3 Methodology

## 3.1 Dataset: BANKING77

The BANKING77 dataset (Casanueva et al., 2020) was developed by PolyAI and consists of 13,083 customer service queries in the banking domain, annotated with 77 intent categories. The dataset was collected to represent the kind of utterances a user might send to a banking chatbot, covering a wide range of topics including account management, card issues, payments, transfers, and exchange rates.

Table 1: BANKING77 dataset statistics.

| **Split** | **Samples** | **Classes** |
|---|---|---|
| Train | 10,003 | 77 |
| Test | 3,080 | 77 |
| Total | 13,083 | 77 |
| Samples/class (avg) | 130 | — |
| Samples/class (min) | 10 | — |
| Samples/class (max) | 170 | — |

The dataset exhibits a near-uniform class distribution, with each of the 77 intents containing between 10 and 170 training examples. The queries are short, averaging approximately 11 words per utterance. Each query belongs to exactly one intent category, making this a single-label, multi-class classification problem.

The dataset is publicly available through the official PolyAI GitHub repository and was loaded directly from raw CSV files hosted at `https://github.com/PolyAI-LDN/task-specific-datasets`.

## 3.2 Data Preprocessing

All text was subjected to a standardized preprocessing pipeline prior to feature extraction or tokenization:

1. **Lowercasing:** all characters were converted to lowercase.
2. **Noise removal:** non-alphabetic characters (punctuation, digits, special characters) were replaced with whitespace using a regular expression (`[^a-z\s]`).
3. **Whitespace normalization:** multiple consecutive spaces were collapsed into a single space, and leading/trailing whitespace was stripped.

This pipeline was applied uniformly to both the training and test sets. Explicit tokenization was intentionally omitted at the preprocessing stage for three reasons: (1) `TfidfVectorizer` performs its own internal tokenization during feature extraction; (2) the BiLSTM pipeline applies whitespace-based splitting at the sequence encoding step, which is compatible with the GloVe vocabulary (GloVe tokens are themselves space-delimited); and (3) the DistilBERT tokenizer applies WordPiece subword segmentation internally, and pre-tokenizing the input prior to this step can interfere with the expected subword splits, as noted in the HuggingFace Transformers documentation (Wolf et al., 2020).

## 3.3 Feature Extraction

**TF-IDF (Traditional ML):** For Logistic Regression, SVM, and Random Forest, text features were extracted using a Term Frequency–Inverse Document Frequency (TF-IDF) vectorizer with the following configuration: unigrams and bigrams (ngram_range = (1,2)), a maximum vocabulary of 30,000 features, and sublinear TF scaling. This configuration was fixed across all three classical models to ensure fair comparison.

**BiLSTM — GloVe tokenization and embeddings:** Words were segmented by whitespace and mapped to integer indices using a vocabulary constructed from all unique tokens in the training corpus. Pre-trained GloVe 6B 100-dimensional vectors (Pennington et al., 2014) were used to initialize the embedding layer. For each word in the vocabulary, its GloVe vector was loaded directly into the corresponding row of the embedding matrix. Words not present in GloVe (out-of-vocabulary tokens, primarily rare or domain-specific terms) were initialized with small random normal values ($\sigma = 0.1$). Sequences were padded with a `<PAD>` token (index 0) or truncated to a maximum length of 40 tokens. The embedding weights were kept trainable during fine-tuning, allowing the pre-trained vectors to adapt to the banking domain. GloVe coverage of the corpus vocabulary exceeded 95%, substantially reducing the out-of-vocabulary rate compared to the previous frequency-based vocabulary approach.

**DistilBERT tokenization:** For the transformer model, text was tokenized using the `distilbert-base-uncased` WordPiece tokenizer, with sequences padded or truncated to a maximum length of 64 tokens.

## 3.4 Models

### 3.4.1 Logistic Regression

A multinomial Logistic Regression with L2 regularization ($C = 1.0$), trained using the L-BFGS solver with a maximum of 1,000 iterations. Implemented using `scikit-learn`.

### 3.4.2 Support Vector Machine (LinearSVC)

A Linear Support Vector Classifier with $C = 1.0$ and a maximum of 3,000 iterations. LinearSVC is particularly effective for high-dimensional sparse feature spaces such as TF-IDF representations (Joachims, 1998).

### 3.4.3 Random Forest

An ensemble of 200 decision trees with bootstrap sampling and the Gini impurity criterion. While Random Forest is less common for text classification due to poor performance on sparse inputs, it was included as a representative tree-based baseline.

### 3.4.4 Bidirectional LSTM with GloVe Embeddings and Attention

A two-layer Bidirectional LSTM with hidden dimension 256, preceded by a GloVe-initialized embedding layer (100-dimensional) and followed by an additive attention mechanism (Bahdanau et al., 2015) that computes a weighted sum over all hidden states. The architecture includes spatial dropout (p=0.3) after the embedding layer, explicit dropout (p=0.3) between the two LSTM layers, Layer Normalization after the second LSTM, and a two-layer fully connected classifier (256 $\rightarrow$ 77 classes) with dropout (p=0.4). Training used AdamW ($lr = 10^{-3}$, weight_decay=$10^{-4}$) with label smoothing ($\epsilon = 0.1$), gradient clipping (max norm = 1.0), and a ReduceLROnPlateau scheduler (patience=3). Early stopping was applied with patience of 8 epochs over a maximum of 50 epochs. The GloVe embedding weights were fine-tuned during training rather than frozen.

### 3.4.5 DistilBERT (fine-tuned)

The `distilbert-base-uncased` model (Sanh et al., 2019) was fine-tuned by appending a linear classification head with 77 output units. Training used AdamW ($lr = 2 \times 10^{-5}$, weight_decay=0.01) with a linear learning rate schedule including 10% warmup steps. The model was trained for 20 epochs with a batch size of 32, with the best checkpoint selected by validation accuracy.

## 3.5 Evaluation Metrics

All models were evaluated on the held-out test set using the following metrics: **Accuracy**, **Macro-averaged Precision**, **Macro-averaged Recall**, **Macro F1-score**, and **Weighted F1-score**. Macro averaging was preferred as the primary metric to give equal weight to all 77 intent classes regardless of class frequency.

# 4 Experiments and Results

## 4.1 Overall Performance Comparison

Table 2 presents the test set performance of all five models across all evaluation metrics.

Table 2: Test set performance of all models on BANKING77.

| Model | Accuracy | Precision | Recall | F1 Macro | F1 Weighted |
|---|---|---|---|---|---|
| Logistic Regression | 0.856 | 0.867 | 0.856 | 0.854 | 0.854 |
| SVM (LinearSVC) | 0.894 | 0.898 | 0.894 | 0.894 | 0.894 |
| Random Forest | 0.866 | 0.871 | 0.866 | 0.865 | 0.865 |
| BiLSTM + GloVe (fine-tuned) | 0.899 | 0.903 | 0.899 | 0.899 | 0.899 |
| DistilBERT (fine-tuned) | **0.932** | **0.934** | **0.932** | **0.932** | **0.932** |

DistilBERT achieves the best performance across all metrics with a Macro F1 of 0.932. The Bidirectional LSTM with GloVe embeddings reaches F1 Macro of 0.899, surpassing SVM (LinearSVC) at 0.894 and improving over the previously reported random-initialization baseline. Logistic Regression (F1 Macro: 0.854) and Random Forest (F1 Macro: 0.865) represent the lower end of the performance range.

Figure 1 illustrates the Accuracy and F1 Macro scores visually.

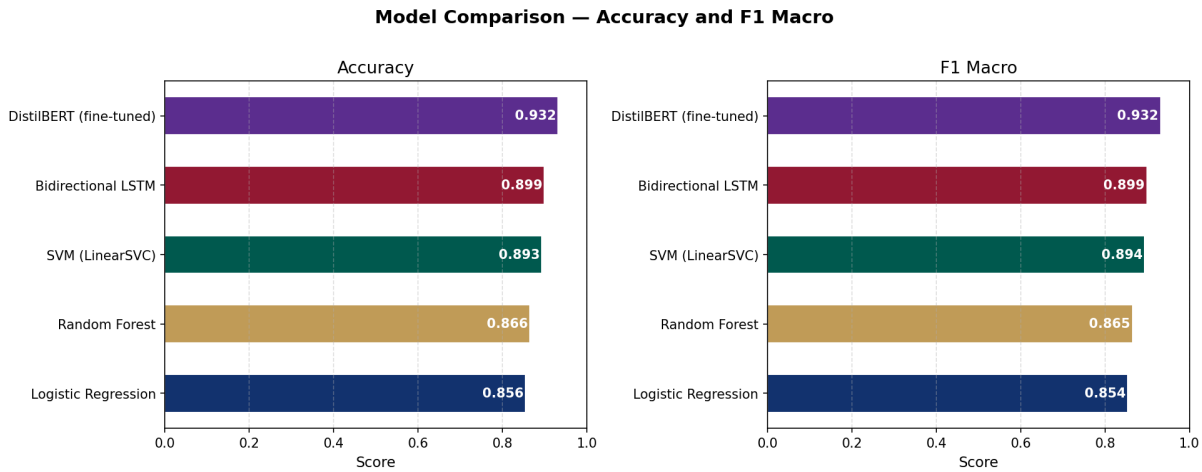


Figure 1: Horizontal bar charts comparing Accuracy (left) and F1 Macro (right) across all five models.

Figure 2 provides a complete view of all metrics simultaneously through a heatmap visualization.

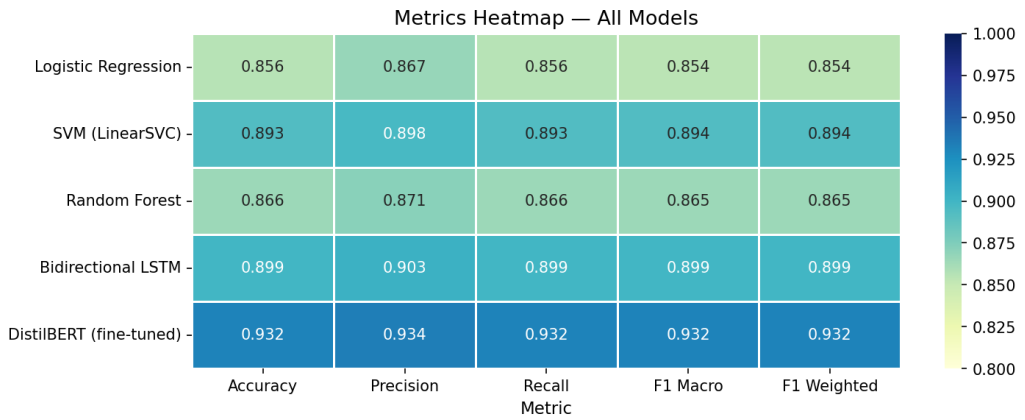


Figure 2: Heatmap of all evaluation metrics for each model. Darker cells indicate higher scores.

Figure 3 shows the radar chart, which makes the overall performance profile of each model visually comparable across all five metrics simultaneously.

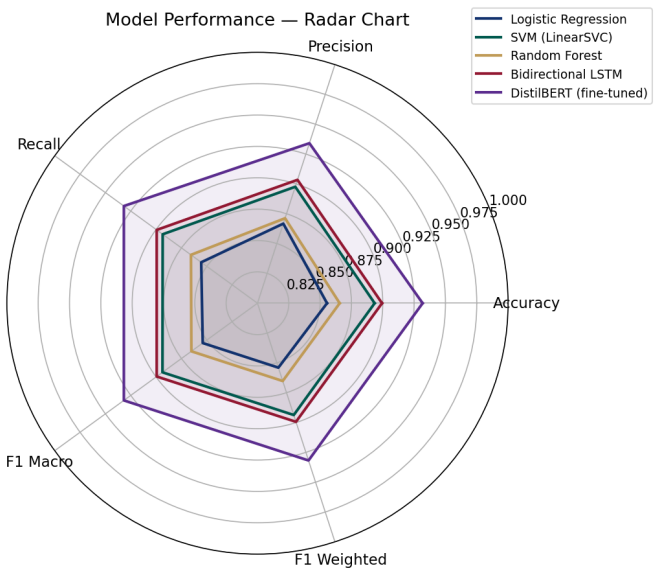


Figure 3: Radar chart of model performance across all five evaluation metrics.

## 4.2 Precision–Recall Analysis

Figure 4 presents a scatter plot of Precision vs. Recall for each model. All models cluster near the diagonal, indicating that precision and recall are well-balanced, which is expected given the near-uniform class distribution in BANKING77.

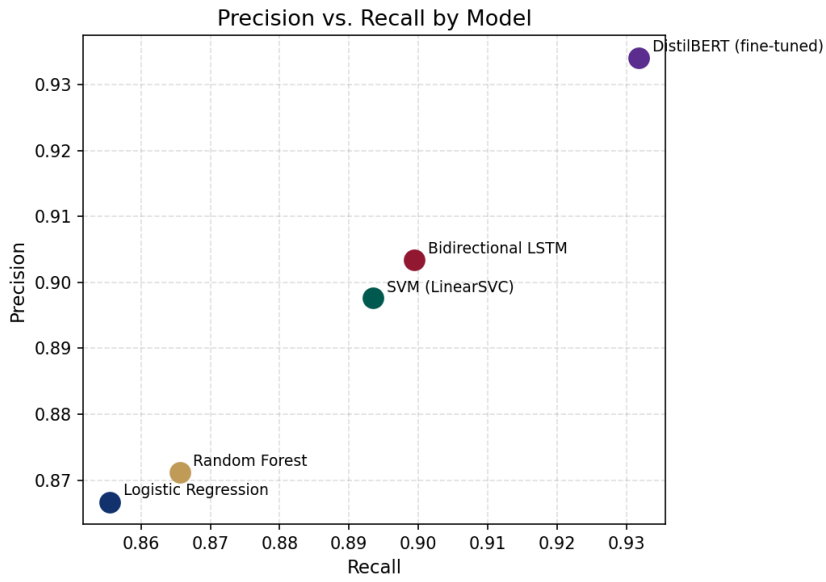


Figure 4: Precision vs. Recall scatter plot. All models exhibit balanced precision and recall, with DistilBERT positioned in the upper-right corner.

## 4.3 Relative Gain over Baseline

Figure 5 quantifies the improvement in F1 Macro of each model relative to Logistic Regression as the baseline. The BiLSTM with GloVe achieves approximately +5.3% relative improvement — surpassing the SVM at +4.7% — while DistilBERT achieves +9.1%.

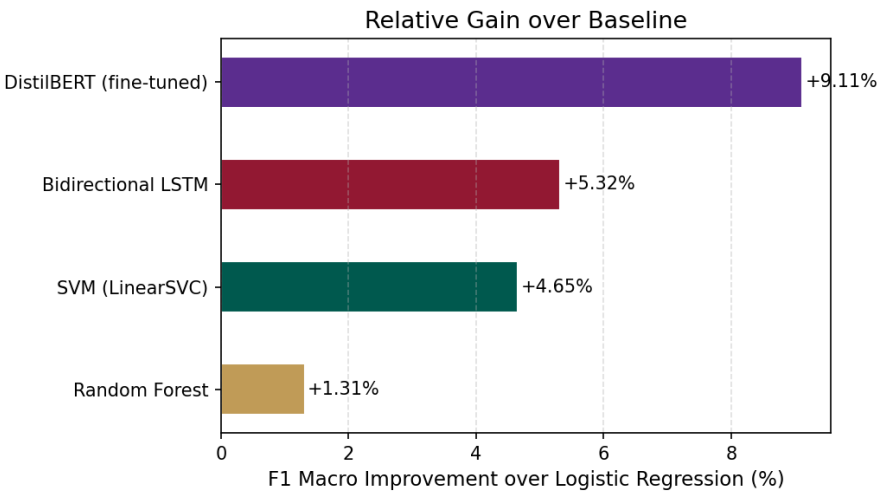


Figure 5: Relative F1 Macro improvement of each model over Logistic Regression as baseline.

## 4.4 Training Dynamics

Figure 6 shows the training curves for both deep learning models. The BiLSTM with GloVe converges smoothly across 30 epochs (early stopping triggered), with validation

accuracy plateauing around 0.909. The faster initial convergence compared to random initialization is consistent with the benefit of a well-initialized embedding layer. The DistilBERT model converges rapidly within the first 5 epochs and reaches peak validation accuracy of 0.932 at epoch 17, with no significant overfitting observed.

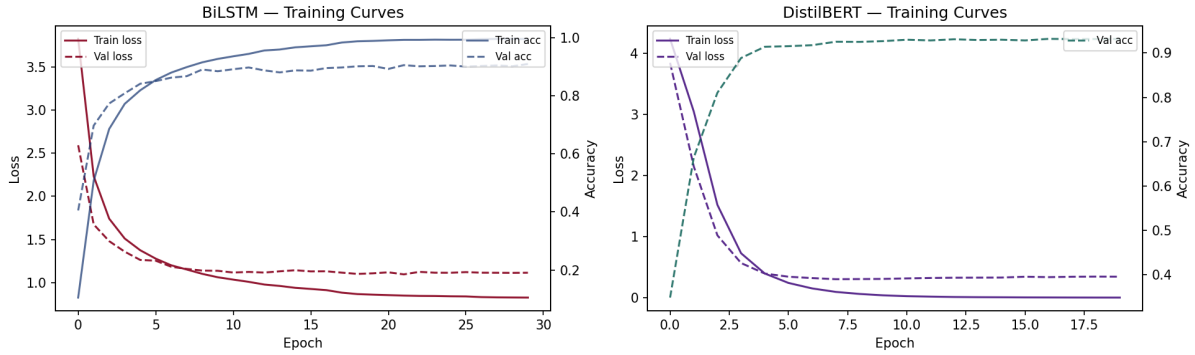


Figure 6: Training and validation loss/accuracy curves for BiLSTM with GloVe (left) and DistilBERT (right).

## 4.5 Per-class Analysis

Figure 7 shows the 15 intent categories with the lowest F1 scores under DistilBERT, while Figure 8 shows the 15 with the highest.

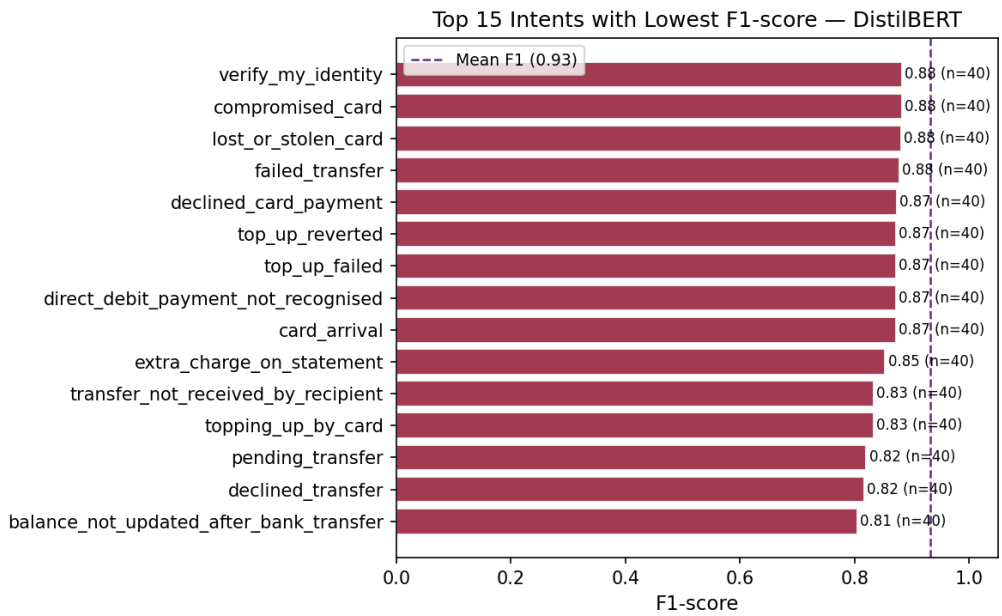


Figure 7: Top 15 intent categories with the lowest F1-score under DistilBERT. The dashed line marks the mean F1 across all 77 classes.

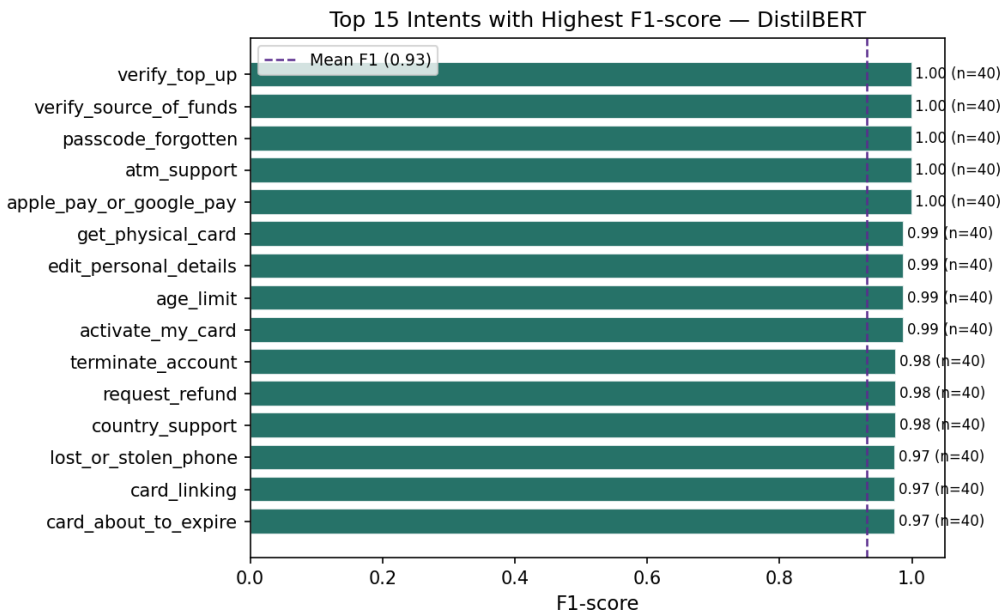


Figure 8: Top 15 intent categories with the highest F1-score under DistilBERT. Several intents achieve a perfect score of 1.00.

The highest-performing intents — including *apple_pay_or_google_pay*, *atm_support*, and *passcode_forgotten* — are lexically distinctive and rarely confused with other categories. The lowest-performing intents tend to share overlapping vocabulary with semantically similar categories, such as pairs involving different types of card or transfer issues.

# 5 Discussion

## 5.1 Model Performance Analysis

The results confirm the general pattern observed in the NLP literature: pre-trained representations outperform models trained from scratch. DistilBERT's advantage (F1 Macro: 0.932) stems from its contextual token representations, which capture semantic nuance that neither TF-IDF nor static word vectors can fully encode. For a 77-class task where many intents are lexically similar, this contextual understanding is decisive.

The BiLSTM with GloVe embeddings (F1 Macro: 0.899) surpasses SVM (0.894), confirming that pre-trained static embeddings provide a measurable benefit even when fine-tuned during training. The GloVe coverage of the corpus vocabulary exceeded 95%, meaning the model starts with meaningful representations for nearly all tokens it encounters in BANKING77.

SVM (LinearSVC) remains a strong competitor: it achieves F1 Macro of 0.894 with no

GPU requirement and training time under one minute, making it highly practical for resource-constrained settings.

Random Forest underperforms relative to SVM and Logistic Regression, which is expected given that tree-based ensembles are generally less effective on high-dimensional sparse representations such as TF-IDF.

## 5.2 Impact of GloVe Initialization

Comparing the BiLSTM with random initialization (F1 Macro: 0.894, previous run) to the BiLSTM with GloVe initialization (F1 Macro: 0.899) reveals a modest but consistent improvement of approximately +0.5 F1 points. This gain is achieved without any architectural change — the only difference is the source of the initial embedding weights. The result is consistent with the broader finding in the NLP literature that pre-trained embeddings provide a useful inductive bias even in tasks with domain-specific vocabulary (Pennington et al., 2014).

## 5.3 Trade-off: Performance vs. Complexity

The relative gain analysis (Figure 5) reveals a clear performance hierarchy with increasing model complexity: Logistic Regression (baseline) $<$ Random Forest $<$ SVM $\approx$ BiLSTM (random) $<$ BiLSTM + GloVe $<$ DistilBERT. SVM provides an excellent cost-benefit ratio at +4.7% over baseline with minimal resources. The BiLSTM with GloVe adds +0.5% over SVM at the cost of GPU training. DistilBERT's further +3.3% gain requires fine-tuning a 66M-parameter transformer.

## 5.4 Error Analysis

The per-class analysis reveals two types of failure modes in DistilBERT. First, intents involving *transfers* — such as *cancel_transfer* and *balance_not_updated_after_bank_transfer* — are frequently confused due to shared vocabulary (*transfer*, *balance*, *bank*). Second, intents involving *exchange rates* and *cash withdrawals* co-occur with similar monetary terminology, making them hard to distinguish. Intents with perfect F1 (1.00) are those with highly distinctive vocabulary such as *apple_pay_or_google_pay* and *passcode_forgotten*.

## 5.5 Strengths and Limitations

**Strengths:** (1) All five models were evaluated on an identical train/test split, ensuring fair comparison. (2) The use of GloVe embeddings is ablated against random initialization, providing direct evidence of the embedding contribution. (3) Both classical and neural

families are represented, enabling a broad evaluation of the performance–complexity trade-off.

**Limitations:** (1) The BiLSTM uses GloVe 6B 100d; larger GloVe variants (200d, 300d) or domain-adapted embeddings may yield additional gains. (2) No hyperparameter search was performed for the classical models beyond default settings. (3) The analysis is limited to a single dataset; cross-domain generalization is not assessed.

# 6 Conclusion

This paper presented a systematic comparison of five machine learning models for banking intent classification on the BANKING77 benchmark. DistilBERT achieved the best overall performance (Accuracy: 0.932, F1 Macro: 0.932), demonstrating the power of pre-trained contextual representations for fine-grained, 77-class intent detection. The Bidirectional LSTM with GloVe pre-trained embeddings (F1 Macro: 0.899) emerged as the best non-transformer model, outperforming both the SVM baseline and a random-initialization BiLSTM variant, and confirming that static pre-trained word vectors provide measurable benefits in recurrent architectures.

Per-class analysis revealed that semantically similar intents sharing overlapping vocabulary — particularly in the domains of transfers and exchange rates — represent the main source of classification errors even for the best model. Future work should explore: (1) larger or domain-adapted GloVe/FastText embeddings for the BiLSTM; (2) data augmentation for low-resource intents; (3) ensemble methods combining transformer and classical model predictions; and (4) error-driven fine-tuning on the hardest intent pairs.

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